
Probe-Guided Gradient Balancing for Multimodal Learning

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Jointly trained multimodal networks frequently underutilize their weaker modalities, as faster-learning modalities dominate the shared fusion objective and suppress the gradient signal reaching slower ones. Most existing approaches throttle the dominant modality, whereas directly boosting the weaker modality often fails because monitoring signals derived from the joint loss are entangled with, and thus dominated by, the stronger modality. We propose a new gradient modulation approach, probe-guided gradient boosting (PGGB), to address the problem by probing and boosting the weak modalities while preserving the strength of the dominant modality. Probing is performed by attaching a lightweight linear classifier to each modality’s stop-gradient features to estimate the corresponding representation quality score. The unbiased gap between these scores across modalities, the *utilization gap*, serves as an imbalance signal that defines a bounded and smoothed scaling factor adaptively reweighing the gradients of weaker modalities, enabling stable and effective rebalancing during joint optimization. The method is composable with throttling methods as the intervention leaves the loss and fusion architecture unchanged. Extensive experiments were conducted across eight benchmarks spanning four domains and 2–4 modalities. The results show that our method outperforms state-of-the-art approaches on highly imbalanced multimodal datasets, while remaining competitive on benchmarks with low or no modality imbalance. Bounded scaling, self-attenuation, and a standard-SGD descent bound were established under standard smoothness and bounded-variance assumptions. Code is provided in the supplementary material.

1 Introduction

24 Multimodal learning integrates complementary signals across heterogeneous sources, including audio, visual, and textual streams among others [4]. Under mild assumptions on latent structure, multimodal models are provably superior to their unimodal counterparts [17]. Despite this theoretical promise, jointly trained multimodal networks frequently fail to fully exploit all available modalities, often performing no better than their strongest unimodal component. This *modality laziness* effect [28, 24, 8] has been observed for instance in audio-dominant emotion recognition on the CREMA-D [6] dataset.

31 Existing methods share a *throttle-only bias*, intervening primarily by damping the dominant modality’s gradient [24, 32, 14, 33, 9, 20]. Some include nominally bidirectional terms [21, 11, 16, 31], but throttling empirically dominates, and Wei et al. [33]’s symmetric boost-direction variant fails to close the gap. In late-fusion, every encoder’s gradient flows through a shared classifier whose loss-derivative is dominated by the strong modality’s contribution to the fusion loss. Amplifying the weak encoder’s gradient therefore rescales an already-corrupted shared signal. Recent methods take partial steps [35, 19] but continue to monitor imbalance through signals coupled to the joint training objective.

We propose *probe-guided gradient boosting* (PGGB): lightweight linear probes [1] on stop-gradient features (the stop-gradient operator passes activations forward but blocks the gradient in the backward pass [27]) yield a per-modality representation-quality estimate, whose cross-modality spread defines the *utilization gap*, independent of the shared classifier’s loss-derivative, removing the chain-rule coupling that impaired prior boost-the-weak attempts [33]. The signal drives a bounded multiplicative factor on the weaker encoder’s gradient, smoothed by an exponential moving average (EMA) [25]. Because the factor acts only on encoder gradients and modifies neither the loss nor the fusion architecture, it is *composable* with throttling methods, multiplying with their suppression factors to produce two-sided pressure toward balance.

We evaluate PGGB on eight benchmarks spanning audio-visual, sentiment, text-image, and medical-segmentation tasks (details in Section 4.1). The main gain is observed on the high-imbalance CREMA-D benchmark (3-frame), where composing PGGB with a representative gradient-modulation baseline [24] reaches $71.45 \pm 1.71\%$ accuracy, +2.31 pp over the same baseline alone and +9.86 pp over joint training. On the four low-imbalance benchmarks (AVE, Kinetics-Sounds, Twitter15, Sarcasm), PGGB alone is the best-performing method on all four (Table 1). On the high-imbalance text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI), where text alone is near-multimodally sufficient and prior balancing methods report only modest gains [21, 30], PGGB+OGM-GE remains within seed standard deviation of the strongest baseline (−0.08 and −0.04 pp). The OGM-GE composition variant has two isolated regressions versus joint training (Kinetics-Sounds −1.72 pp, Sarcasm −0.64 pp) noted in Section 5, while several gradient-modulation baselines degrade on multiple datasets (e.g., this baseline [24] regresses on Kinetics-Sounds by −1.80 pp, Table 1).

Contributions.

- We identify a *throttle-only bias* in the modality-balancing literature and show that its natural complement, boosting the weaker modality, fails when the monitoring signal is coupled to the joint training objective via shared loss derivatives in late fusion.
- We propose *decoupled probe monitoring*, a diagnostic attaching lightweight linear probes to stop-gradient features, yielding a unbiased *utilization gap* estimate (the spread of per-modality probe accuracies) independent of the joint objective. The decoupling enables effective weak-modality boosting where prior coupled-monitoring boost-the-weak attempts failed [33].
- We introduce *PGGB*, one instantiation that converts the monitoring signal into a bounded multiplicative scaling on the weaker encoder’s gradients. It is composable with existing throttling methods (Appendix B.7 validates non-destructive composition with four such methods).
- We validate *PGGB* on eight benchmarks spanning audio-visual, sentiment, text-image, and medical-segmentation tasks. *PGGB* composed with throttling outperforms state-of-the-art baselines on high-imbalance audio-visual (CREMA-D), matches the strongest available baseline on medical segmentation (BraTS 2021, +0.28 pp, within seed variance), and remains competitive within seed variance on the high-imbalance text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI), and *PGGB* used alone is the best-performing method on all four low-imbalance datasets (AVE, Kinetics-Sounds, Twitter15, Sarcasm).

2 Related work

Modality imbalance in joint training. Jointly trained multimodal networks frequently underperform their best unimodal component [28]. Under a Gaussian mixture model, Huang et al. [18] prove that joint training learns only a subset of modalities and derive bounds on the duration of the resulting unimodal phase, and Du et al. [8] characterize the conditions under which individual modality features fail to be learned. Zhang et al. [44] further show, in deep linear networks, that this unimodal phase depends on architecture depth and initialization. Modality imbalance is therefore a structural property of joint optimization, rooted in the shared loss through which all encoder gradients flow.

Gradient modulation for modality balance. The dominant response has been to modulate gradient magnitudes during backpropagation. Multi-task precursors GradNorm [7] and PCGrad [39] inspired

a line of multimodal work. OGM-GE [24] dampens the dominant modality’s gradient via a ratio of per-modality contributions to the target-class score, refined by AGM [21], CGGM [14], and MLGM [20]. Although some methods include bidirectional terms, throttling empirically dominates and boost-only variants remain ineffective: the symmetric boost direction (OGM*) reported in Wei et al. [33] fails to close the modality gap, which we attribute to coupled monitoring signals affected by the dominant modality (Section 3). M-SAM [23] pairs Shapley attribution with sharpness-aware perturbation, inheriting the same coupled-monitoring limitation.

Alternative approaches to modality balance. Beyond gradient scaling, a broader set of interventions has been proposed. These include feed-forward feature modulation [33] and Pareto-optimal aggregation [32]. Modality-alternating schemes interleave unimodal and joint updates [15, 43]. Diagnostic methods detect imbalance through separability [34], Fisher information [16], or bias-variance decomposition [31]. Other approaches intervene at the data level [22, 37]. Per-modality learning-rate methods [36, 13] are an adjacent class operating on the optimizer step rather than the gradient. Despite their diversity, these methods share a common design choice. Imbalance is detected through signals *coupled* to the joint objective, such as gradient ratios, per-modality losses, or live-model predictions (with the partial exception of PMR’s decoupled unimodal prototypes [9]). These are precisely the quantities distorted by dominant-modality dynamics, which limits the reliability of the resulting corrections.

Representation monitoring and probing. Linear probes on frozen representations [1] are a standard tool for assessing representation quality. Several multimodal methods adopt probe-like monitoring but retain coupling to the joint objective. IPRM’s two-pass forward strategy compares unimodal and fused predictions of the live model via KL divergence [35], CGGM’s auxiliary classifiers participate in the training loss [14], and AUG [19] relies on loss-based weakness signals that inherit the dominant modality’s bias. PMR [9] moves toward decoupling via unimodal prototypes but the prototype signal feeds a gradient coefficient on the shared encoder, while DI-MML [10] detaches only the unimodal pre-fusion stage. In contrast, we train *fully decoupled* linear probes on stop-gradient features with separate optimizers and evaluate them on held-out within-batch samples to prevent probe overfitting. The resulting utilization gap directly boosts the weaker encoder’s gradients, so the imbalance estimate is not affected by the dynamics it aims to correct.

3 Probe-Guided Gradient Boosting

We present PGGB in four stages: problem formulation (§3.1), decoupled probe monitoring (§3.2), the boost mechanism (§3.3), and composability with existing balancing methods (§3.4). Figure 1 traces the data flow: the top panel depicts the forward path (§3.1), the bottom panel depicts probe monitoring on stop-gradient features (§3.2), and the green boost block plus actuation rail depict the gradient-modification mechanism (§3.3). Composition with throttling is described in (§3.4).

3.1 Problem formulation

Each multimodal sample comprises M inputs $\{\mathbf{x}_m\}_{m=1}^M$, one per modality. Encoder f_m with parameters θ_m maps $\mathbf{x}_m$ to $\mathbf{z}_m$:

$$\mathbf{z}_m = f_m(\mathbf{x}_m; \theta_m) \in \mathbb{R}^d. \quad (1)$$

Each $\mathbf{z}_m$ is a real value feature vector in $\mathbb{R}^d$, with d common across modalities.

$$\mathcal{L}_{\text{fusion}} = \ell(\hat{y}, y). \quad (2)$$

Here ℓ is a differentiable per-sample loss mapping a prediction-target pair to a non-negative scalar. By the chain rule, the gradient with respect to encoder parameters decomposes as

$$\partial \mathcal{L}_{\text{fusion}} / \partial \theta_m = (\partial \mathcal{L}_{\text{fusion}} / \partial g) (\partial g / \partial \mathbf{z}_m) (\partial \mathbf{z}_m / \partial \theta_m). \quad (3)$$

The term $\partial \mathcal{L}_{\text{fusion}} / \partial g$ is shared across all modalities, since all encoder outputs flow through the same fusion head. This shared dependency is the root of coupled monitoring failure: when one modality (say $m=1$) converges faster and begins to dominate the head’s outputs, $\partial \mathcal{L}_{\text{fusion}} / \partial g$ increasingly reflects the dominant modality’s contribution. Any monitoring signal derived from $\mathcal{L}_{\text{fusion}}$ or its gradients, including the discrepancy ratio used by OGM-GE [24] and the auxiliary classifier losses used by CGGM [14], inherits this bias.

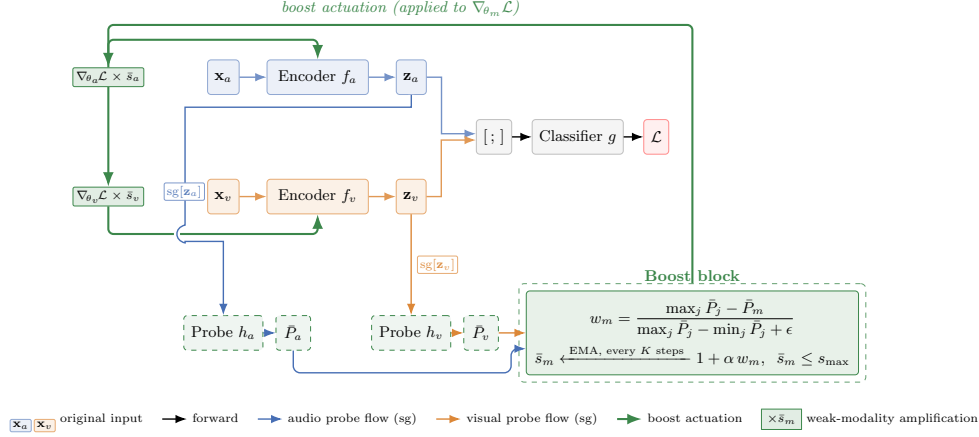


Figure 1: An Illustration of probe-guided gradient boosting on 2 modalities. **Top panel:** features $\mathbf{z}_m$ are concatenated and classified via the shared fusion classifier g , producing $\mathcal{L}$. **Bottom panel:** stop-gradient features $\text{sg}[\mathbf{z}_m]$ are evaluated by linear probes along solid per-modality probe-flow lines (audio in blue, visual in orange), yielding per-modality accuracies $\bar{P}_m$. The boost block computes the relative-gap weight w_m and a smoothed scale $\bar{s}_m$ (EMA-smoothed, refreshed only at probe-evaluation events every K steps and held constant in between), clipped at $s_{\max}$. The actuation rail (thick green) applies $\bar{s}_m$ to the per-modality gradient $\nabla_{\theta_m} \mathcal{L}$. Probes never backpropagate into encoders, so the monitoring signal is structurally decoupled from the fusion objective.

3.2 Decoupled probe monitoring

We attach a lightweight linear probe classifier $h_m: \mathbb{R}^d \rightarrow \mathbb{R}^C$ (with C outputs) to each encoder, parameterized by ψ_m . The resulting probe accuracies define an imbalance signal, the *utilization gap* (Eq. 6), independent of the joint training objective. The probes are trained on stop-gradient features $\text{sg}[\mathbf{z}_m]$, where $\text{sg}[\cdot]$ denotes the stop-gradient operator [27] (identity at the forward pass, zero partial derivatives at the backward pass):

$$\mathcal{L}_{\text{probe},m} = \ell_{\text{probe}}(h_m(\text{sg}[\mathbf{z}_m]; \psi_m), y). \quad (4)$$

Each probe has its own optimizer that updates only ψ_m , and no gradient flows into θ_m . Since $\text{sg}[\mathbf{z}_m]$ halts backward propagation at the feature level, $\mathcal{L}_{\text{probe},m}$ depends only on the quality of $\mathbf{z}_m$ as a representation, not on how it contributes through g . This is precisely why decoupling matters: the probe accuracy P_m measures representation quality not influenced by the dominant modality as the influence can be both ways.

Split-batch evaluation and on-the-fly estimation. Every K training iteration, we partition the (shuffled) mini-batch into two disjoint halves, training probes on the first half and evaluating on the second. This yields a unbiased estimate P_m of the current probe’s training-distribution accuracy (conditioned on $\psi_m^{(t)}$), the evaluation half is held out from probe training and drawn i.i.d. from the training distribution. To convert this per-evaluation sample into a stable estimator, we accumulate an exponential moving average (EMA):

$$\bar{P}_m \leftarrow \beta \cdot P_m + (1 - \beta) \cdot \bar{P}_m, \quad (5)$$

where $\beta \in [0, 1]$ is the smoothing coefficient and other low-pass smoothing methods apply. At $\beta = 0.1$ (§4.1), the steady-state variance is $\beta/(2 - \beta) \sigma^2 \approx 0.053 \sigma^2$ under an i.i.d. assumption with per-sample variance σ^2 , approximately $19\times$ smaller than a single-batch estimate. The *utilization gap* is the spread of smoothed probe accuracies:

$$\delta = \max_m \bar{P}_m - \min_m \bar{P}_m. \quad (6)$$

Appendix B.6 verifies empirically that the EMA estimator substantially reduces mini-batch noise in the probe signal during training.

3.3 Probe-guided gradient boosting

Given the EMA-smoothed probe accuracies $\{\bar{P}_m\}_{m=1}^M$ (Eq. 5), we compute a per-modality boost scale that amplifies the gradients of weaker encoders (full pseudocode in Appendix A.1, Algorithm 1). The procedure involves three steps.

Step 1: Relative weakness. For each modality m , we compute a normalized weakness score $w_m \in [0, 1]$ (zero for the strongest modality, one for the weakest):

$$w_m = \frac{\max_j \bar{P}_j - \bar{P}_m}{\max_j \bar{P}_j - \min_j \bar{P}_j + \epsilon}, \quad (7)$$

where j ranges over all modalities $\{1, \dots, M\}$, and ϵ is a small positive constant serving as a numerical guard against division by zero.

Step 2: Boost scale. w_m is converted to a multiplicative scaling factor:

$$s_m = \min(1 + \alpha \cdot w_m, s_{\max}), \quad (8)$$

where $\alpha > 0$ controls the boost strength and $s_{\max} > 1$ caps the maximum amplification to prevent extreme scaling. When modalities are balanced (δ small), weakness scores become near zero and $s_m \approx 1$, so the method applies no intervention (Proposition 2 gives the precise continuity bound).

Step 3: Smoothing and gradient modification. To avoid abrupt changes in the boost signal, we apply temporal smoothing to the scales. We use a separate exponential moving average (EMA), distinct from the probe-accuracy EMA in Eq. (5), though our analysis (Propositions 2 and 3) is stated specifically for this EMA:

$$\bar{s}_m \leftarrow \mu \cdot s_m + (1 - \mu) \cdot \bar{s}_m, \quad (9)$$

where $\mu \in (0, 1)$ is the EMA smoothing coefficient on the boost scales (set in Section 4). The smoothed scales are applied to the encoder gradients *after* the backward pass and *before* the optimizer step:

$$\frac{\partial \mathcal{L}}{\partial \theta_m} \leftarrow \bar{s}_m \cdot \frac{\partial \mathcal{L}}{\partial \theta_m}. \quad (10)$$

3.4 Composability with gradient throttling

Because PGGB amplifies weaker modalities without modifying the dominant modality’s gradient, it is mechanistically compatible with synchronous balancing methods such as gradient throttling, loss reweighting, and Pareto rescaling, with the empirical magnitude of the compound gain depending on the specific base method (Section 4.2, Appendix B.7). The main experiments compose with OGM-GE [24], and Appendix B.7 validates composition with four additional methods.

OGM-GE damps the dominant modality’s gradient by a *coupled* signal (inter-modal gradient-magnitude ratio ρ derived from $\mathcal{L}_{\text{fusion}}$), whereas our boost uses a *decoupled* signal (probe accuracy on stop-gradient features $\text{sg}[\mathbf{z}_m]$). The two mechanisms compose multiplicatively on encoder gradients:

$$\frac{\partial \mathcal{L}}{\partial \theta_m} \leftarrow \bar{s}_m \cdot \omega_m^{\text{ogm}} \cdot \frac{\partial \mathcal{L}}{\partial \theta_m}, \quad (11)$$

where $\omega_m^{\text{ogm}} \leq 1$ is the OGM-GE throttle and $\bar{s}_m \geq 1$ is our boost scale.

Following established practice for auxiliary unimodal supervision [14], we optionally include unimodal auxiliary losses, yielding the total objective

$$\mathcal{L} = \mathcal{L}_{\text{fusion}} + \gamma \sum_{m=1}^M \ell_{\text{aux}}(g_m(\mathbf{z}_m), y), \quad (12)$$

where γ is a scalar weight on the per-modality auxiliary terms (set in Section 4). ℓ_{aux} matches the form of ℓ (Eq. 2) for each benchmark.

3.5 Safety properties

We establish three safety properties: bounded scaling, self-attenuation, and a standard-SGD descent bound. These are structural guarantees that the method does not diverge or harm low-imbalance training, not mechanism-specific convergence results. A rate-level analysis tying gap closure to probe dynamics is left for future work. Full proofs are in Appendix A.

Assumption 1 (Smoothness). *The loss $\mathcal{L}(\theta)$ is L -smooth: $\|\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\theta')\| \leq L\|\theta - \theta'\|$ for all θ, θ' .*

Assumption 2 (Bounded variance). *The stochastic gradient $\mathbf{g}(\theta)$ satisfies $\mathbb{E}[\mathbf{g}(\theta)] = \nabla\mathcal{L}(\theta)$ and $\mathbb{E}[\|\mathbf{g}(\theta) - \nabla\mathcal{L}(\theta)\|^2] \leq \sigma^2$ for all θ .*

Proposition 1 (Bounded scaling). *For all modalities m and all iterations t , the smoothed boost scale satisfies $\bar{s}_m^{(t)} \in [1, s_{\max}]$.*

Proposition 2 (Self-attenuation). *For any $\Delta \geq 0$, if the smoothed probe accuracies are Δ -balanced ($\max_j \bar{P}_j - \min_j \bar{P}_j \leq \Delta$) at iteration t , then each modality’s instantaneous boost scale satisfies $|\bar{s}_m^{(t)} - 1| \leq \alpha\Delta/(\Delta + \epsilon)$, and the EMA-smoothed scale converges geometrically to unity at rate $(1 - \mu)$ as long as this condition persists.*

Proposition 3 (Descent bound). *Under Assumptions 1 and 2, SGD with PGGB and constant learning rate $\eta \leq 1/(L \cdot s_{\max}^2)$ satisfies*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla\mathcal{L}(\theta_t)\|^2] \leq \frac{2\Delta_{\mathcal{L}}}{\eta T} + L\eta s_{\max}^2 \sigma^2, \quad (13)$$

where $\Delta_{\mathcal{L}} = \mathcal{L}(\theta_0) - \mathcal{L}^*$ is the initial optimality gap and $\mathcal{L}^*$ is a lower bound on $\mathcal{L}$.

The bound matches the standard nonconvex SGD rate [12] up to an $s_{\max}^2$ factor in the variance term, a controlled increase ($s_{\max} = 2$), and Proposition 2 ensures this cost is incurred only when imbalance is present. The smoothness constant L in Assumption 1 is loss-dependent, determined by the choice of ℓ in Eq. (2); in our experiments, the per-task learning rate (Section 4) is selected consistent with the descent inequality for the corresponding ℓ .

4 Experiments

4.1 Experimental setup

Datasets. We evaluate PGGB on eight benchmarks spanning three tasks and 2–4 modalities, reported as mean $\pm$ std over five seeds. *Audio-visual*: CREMA-D [6] (3-frame at 3 fps, following OGM-GE), AVE [26], Kinetics-Sounds (KS) [2]. *Sentiment*: CMU-MOSEI [41], CMU-MOSI [40]. *Text-image*: Twitter15 [38, 42], Sarcasm [5]. *Segmentation*: BraTS 2021 [3]. Encoders follow task grouping (ResNet-18 for audio-visual, MLPs on pre-extracted features for sentiment and text-image, DeepLab v3+/ResNet-101 for BraTS), with splits and caveats in Appendix B. We assume each modality is task-relevant, consistent with these benchmarks’ established multimodal use; PGGB does not validate relevance and would amplify a noise-only modality if one were present.

Baselines. We compare experiment results against eight methods: **Baseline** (joint training, no balancing), **OGM-GE** [24] (discrepancy-ratio gradient throttling), **MMPareto** [32] (Pareto-optimal gradient weights), **AGM** [21] (Shapley-based adaptive gradient modulation), **G-Blend** [28] (OG-ratio loss weighting), **CGGM** [14] (classifier-guided magnitude/direction modulation¹), **InfoReg** [16] (unimodal information regularization), and **MILES** [13] (per-modality learning-rate adjustment). InfoReg and MILES are evaluated on CREMA-D only, as their official implementations target audio-visual classification. MMPareto, AGM, and G-Blend are evaluated on all classification benchmarks but not on BraTS 2021, as their official implementations target classification losses and do not support dense-prediction multi-class Dice objectives out of the box.

Reproduction protocol. All baselines are re-run under a matched pipeline (shared encoders, 3-frame CREMA-D, identical augmentation and optimizer) for fair comparison. OGM-GE (69.14%) is

¹Adapted from its original Transformer setting to our CNN/MLP pipeline. The reproduction shows consistent regression across all eight benchmarks, including off-table Food101 (−37 pp vs. baseline, Appendix B), which we attribute to backbone mismatch rather than the underlying method.

3–6 pp below the 72–75% in follow-up work using different frame sampling, AGM (57.42%) uses our ResNet-18 rather than its original backbone, MILES is ported to our matched ResNet-18 pipeline from its specialized backbone, and CGGM from its Transformer setting. Sub-baseline AGM/MILES numbers plausibly reflect 3-frame protocol sensitivity, and published configurations are retained without retuning.

Implementation details. Classification uses concatenation fusion, batch size 64, 100 epochs. Audio-visual datasets use SGD (learning rate 0.001, weight decay 10^{-4}) with StepLR, sentiment and text-image use Adam (learning rate 0.001) with StepLR (step 40), and BraTS uses SGD with cosine scheduling (base learning rate 0.01, 5-epoch warmup) and batch size 12. PGGB uses identical hyperparameters across all datasets: $K=20$, $\beta=0.1$, $\mu=0.3$, $s_{\max}=2.0$, $\gamma=1.0$, and $\alpha \in \{0.5, 0.75\}$ for the standalone and OGM-GE-combined variants respectively. Probes are trained with Adam (learning rate 10^{-3}).

Computational overhead. Each linear probe $h_m: \mathbb{R}^d \rightarrow \mathbb{R}^C$ adds at most a few thousand parameters per modality, negligible relative to the encoders (e.g., ResNet-18 at $\sim 11\text{M}$ parameters). Probe training every $K=20$ iterations on a half mini-batch and the multiplicative gradient scaling at the optimizer step together add approximately 1% wall-clock overhead relative to the baseline. The EMA state $\{\bar{P}_m, \bar{s}_m\}$ requires $O(M)$ additional scalars.

Imbalance categorization. We categorize benchmarks by the utilization gap δ (Eq. (6)) measured during baseline training, with per-dataset values in Appendix B.9: *high-imbalance* ($\delta > 0.15$, CREMA-D, CMU-MOSI, CMU-MOSEI) and *low-imbalance* ($\delta \leq 0.15$, AVE, KS, Twitter15, Sarcasm). BraTS 2021 is reported separately as a dense-prediction segmentation benchmark. The high-imbalance group is heterogeneous in source: CREMA-D imbalance arises from audio-dominance in emotion recognition [24], whereas CMU-MOSI/MOSEI imbalance reflects text-sufficiency, with text-only baselines within roughly 1 pp of multimodal accuracy on CMU-MOSEI [21] and only a +2.7 pp multimodal gain on CMU-MOSI [30]. On these text-dominant benchmarks, balancing methods report only modest absolute gains [14, 33], motivating within-seed-std competitiveness as the relevant metric.

Table 1: Main results across eight benchmarks. We report accuracy (%) for classification datasets and average Dice (%) for BraTS 2021. All values are mean $\pm$ std over 5 seeds. **Bold** indicates the best result per dataset. Imbalance categorization is described in Section 4.1.

Method	CREMA-D	AVE	KS	CMU-MOSI	Twitter15	Sarcasm	CMU-MOSEI	BraTS 2021
Baseline	61.59 $\pm$ 0.80	86.54 $\pm$ 0.42	79.05 $\pm$ 0.40	72.42 $\pm$ 0.48	66.60 $\pm$ 0.57	82.40 $\pm$ 0.17	70.42 $\pm$ 0.29	85.77 $\pm$ 0.62
OGM-GE	69.14 $\pm$ 1.13	86.96 $\pm$ 0.71	77.25 $\pm$ 0.79	72.68 $\pm$ 0.89	66.33 $\pm$ 0.21	81.81 $\pm$ 0.23	72.47 $\pm$ 0.70	86.21 $\pm$ 0.17
MMPareto	65.51 $\pm$ 0.87	86.37 $\pm$ 0.33	78.21 $\pm$ 0.38	72.68 $\pm$ 0.79	66.81 $\pm$ 0.47	82.27 $\pm$ 0.12	70.20 $\pm$ 0.71	—
AGM	57.42 $\pm$ 0.73	84.42 $\pm$ 0.26	77.85 $\pm$ 0.30	72.50 $\pm$ 0.62	66.79 $\pm$ 0.47	82.32 $\pm$ 0.16	69.28 $\pm$ 0.67	—
G-Blend	61.10 $\pm$ 1.87	87.09 $\pm$ 0.29	77.75 $\pm$ 1.13	72.36 $\pm$ 0.66	66.66 $\pm$ 0.57	82.35 $\pm$ 0.26	70.15 $\pm$ 0.66	—
CGGM	50.22 $\pm$ 1.39	76.72 $\pm$ 0.42	73.18 $\pm$ 0.27	59.45 $\pm$ 0.36	62.41 $\pm$ 0.36	80.71 $\pm$ 0.27	68.05 $\pm$ 0.46	81.13 $\pm$ 1.34
InfoReg	67.72 $\pm$ 0.83	—	—	—	—	—	—	—
PGGB ($\alpha=0.5$)	62.72 $\pm$ 1.65	87.41 $\pm$ 0.26	79.17 $\pm$ 0.97	71.89 $\pm$ 0.82	66.85 $\pm$ 0.50	82.44 $\pm$ 0.35	69.80 $\pm$ 0.80	85.98 $\pm$ 1.15
PGGB+OGM-GE ($\alpha=0.75$)	71.45 $\pm$ 1.71	87.23 $\pm$ 0.58	77.33 $\pm$ 0.64	72.60 $\pm$ 0.94	66.59 $\pm$ 0.51	81.76 $\pm$ 0.08	72.43 $\pm$ 0.65	86.49 $\pm$ 0.21

4.2 Main results

Table 1 reports results across eight benchmarks. On high-imbalance CREMA-D, PGGB+OGM-GE attains 71.45%, +2.31 pp over OGM-GE and +9.86 pp over baseline, the best among competitors. A three-way decomposition in §4.3 isolates this gain to the boost actuation, with the decoupled monitor causally inert at $\alpha=0$. PGGB alone (+1.13 pp) is insufficient under severe imbalance, confirming that two-sided intervention is required. On the four low-imbalance benchmarks (AVE, KS, Twitter15, Sarcasm, with cross-class MSLR comparison in Appendix B.10), PGGB alone is the best-performing method on all four, whereas OGM-GE often regresses (KS -1.80 pp, Twitter15 -0.27 pp, Sarcasm -0.59 pp), consistent with self-attenuation as $\delta \rightarrow 0$ (§3.3). On the high-imbalance text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI), where text alone is near-multimodally sufficient and prior balancing methods report only modest absolute gains [21, 30], PGGB+OGM-GE remains within seed standard deviation of the strongest baseline (MOSI -0.08 pp, MOSEI -0.04 pp). On BraTS 2021 (DeepLab v3+/ResNet-101, Dice loss), PGGB+OGM-GE yields 86.49% Dice (+0.72 pp), extending the mechanism to dense prediction. The composability claim further extends on AVE to two of four non-OGM-GE methods tested (Appendix B.7). A comparison against OPM [33], whose single-layer-fusion dependency requires a separate protocol, is in Appendix B.2.

4.3 Ablation studies

We ablate on CREMA-D with single-frame visual input, which widens the modality gap and amplifies method differences (Table 2).

Table 2: Ablation on CREMA-D (1-frame). Accuracy (%), mean $\pm$ std over 5 seeds. The upper row reports the baseline. Lower rows isolate single-mechanism interventions (boost only, OGM-GE only) and their combinations.

Method	Accuracy (%)
Baseline	60.30 $\pm$ 0.70
PGGB ($\alpha=0.5$)	60.46 $\pm$ 0.85
OGM-GE only	62.47 $\pm$ 1.42
PGGB+OGM-GE ($\alpha=0.5$)	62.37 $\pm$ 0.57
PGGB+OGM-GE ($\alpha=0.75$)	62.69 $\pm$ 0.22

Among single-mechanism interventions, OGM-GE alone yields +2.17 pp versus PGGB alone +0.16 pp, consistent with prior findings that throttling is the more impactful individual intervention [33]. Combining PGGB with OGM-GE at $\alpha=0.75$ achieves the best result (62.69%) with a sixfold reduction in std (0.22% vs. 1.42%), indicating that decoupled probe monitoring provides a stabilizing signal complementing the noisier gradient-ratio throttling.

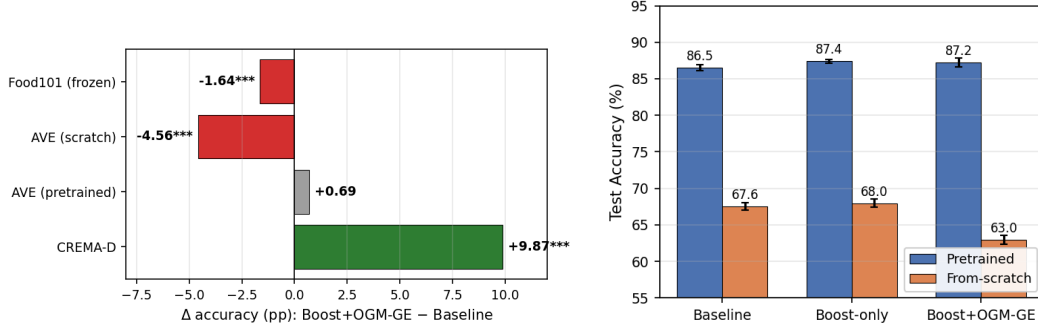
Isolating the monitor from the boost. We run three 5-seed decompositions on CREMA-D 3-frame (Table 1 protocol): (i) $\alpha=0$ with probes active (69.14 $\pm$ 1.13%, reported as OGM-GE alone in Table 1), (ii) an independent 5-seed replication of the same $\alpha=0$ condition (69.14 $\pm$ 1.18%), and (iii) $\alpha=0.75$ (71.45 $\pm$ 1.71%). Conditions (i) and (ii) are operationally identical (identical α , identical probe pipeline) and yield the same mean within 0.00 pp, confirming estimate stability. The +2.31 pp lift at (iii) isolates boost actuation: it cannot be attributed to probe-pipeline side-effects matched across conditions (probe training on stop-gradient features, optimizer steps on probe parameters, and EMA state updates). Only the multiplicative scale s_m differs between $\alpha=0$ and $\alpha=0.75$.

4.3.1 Operating conditions ablation

A complementary question to *which* component drives the CREMA-D gain is *when* the method is effective. We characterize four regimes along three axes (feature learning, encoder initialization, data-to-architecture ratio) in Table 3 and Figure 2.

Table 3: Operating conditions across four regimes. Columns: trainable encoders, modality asymmetry (pretraining or LR differences), adequate data-to-architecture ratio, and complementary modalities. Δ : accuracy change (pp), PGGB+OGM-GE vs. baseline. Significance: *** $p < 0.001$ (Welch’s t-test, 5 seeds per condition). AVE (from-scratch) is the clearest failure case (−4.56 pp), with the pretrained $\leftrightarrow$ scratch contrast localizing sensitivity to the OGM-GE component. Per-regime analysis in Appendix B.3.

Regime	Trainable	Asymmetry	Adequate data	Compl.	Δ
CREMA-D	✓	✓	✓	✓	+9.86***
AVE (pretrained)	✓	✓ (pretrain)	✓	✓	+0.69
AVE (from-scratch)	✓	×	× ($\sim$ 4K)	✓	−4.56***
Food101 (frozen)	×	—	✓	✓	−1.64



(a) Δ accuracy (PGGB+OGM-GE vs. baseline) across four regimes. (b) AVE test accuracy: pretrained vs. from-scratch encoders.

Figure 2: Operating-conditions ablation (5 seeds). (a) PGGB alone self-attenuates toward baseline when Table 3 conditions are violated, whereas PGGB+OGM-GE inherits OGM-GE’s regime-dependent failure modes (AVE from-scratch is the clearest case). (b) With data and architecture fixed, flipping initialization from pretrained to random shifts PGGB+OGM-GE from +0.69 to -4.56 pp ($p=0.0001$) while PGGB alone stays positive, localizing sensitivity to the OGM-GE component.

Encoder initialization and scope. Fixing dataset and architecture but varying encoder initialization gives the sharpest contrast (Figure 2b). PGGB alone stays positive in both regimes (+0.87/+0.42 pp, both $p < 0.05$), while PGGB+OGM-GE flips sign (+0.69 $\rightarrow$ -4.56 pp, $p=0.0001$). ImageNet pretraining installs an asymmetry aligned with OGM-GE’s throttling assumption. Under from-scratch initialization both encoders are simultaneously under-trained on $\sim 4K$ samples, so suppressing either destabilizes the joint dynamic. This localizes the failure to the OGM-GE component (Appendix B.3).

4.4 Analysis

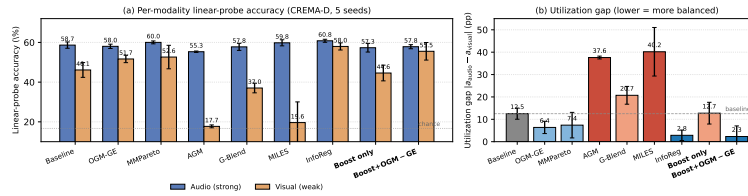


Figure 3: Per-modality diagnostics on CREMA-D (5 seeds). (a) Post-hoc linear-probe accuracy P_m^{post} (Appendix B.8). Dotted line marks chance (1/6). AGM and MILES collapse the weak probe. (b) Utilization gap $|P_a^{\text{post}} - P_v^{\text{post}}|$. PGGB+OGM-GE attains 2.31 pp ($5.4\times$ reduction vs. baseline gap). Error bars: ± 1 std.

We probe per-modality utilization with linear classifiers trained on frozen encoder features. Post-hoc linear probes (Appendix B.8, Figure 3b) show PGGB+OGM-GE attains the smallest utilization gap (2.31 pp): the weak probe rises +9.41 pp, the strong falls only -0.81 pp, while AGM, MILES, and G-Blend degrade the weak probe. Extension to M modalities uses pairwise weakness (Eq. 7).

5 Conclusion

We propose PGGB, a method for balanced multimodal learning that amplifies the contribution of weaker modalities while preserving stronger ones, composing naturally with existing throttling techniques. PGGB alone is the best-performing method on all four low-imbalance benchmarks. On high-imbalance CREMA-D, composition with OGM-GE yields +2.31 pp over the strongest baseline, and on the text-dominant sentiment benchmarks (CMU-MOSI, CMU-MOSEI) the same composition remains within seed standard deviation of the strongest baseline. On medical segmentation (BraTS 2021), it matches the strongest available baseline (+0.28 pp, within seed variance). **Limitations.** Per-setting tuning is required, probes add small compute overhead, and composition with throttling can regress when the dominance gap is small (worst case AVE-from-scratch -4.56 pp, §4.3.1), a regime our theory does not cover. **Future work.** Probe-driven adaptive composition that engages throttling only when the dominance gap warrants, and extension beyond classification and segmentation. No new data is introduced, and §4.3.1 should precede deployment.

References

- [1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations (Workshop)*, 2017.
- [2] Relja Arandjelović and Andrew Zisserman. Look, listen and learn. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 609–617, 2017.
- [3] Ujjwal Baid, Satyam Ghodasara, Suyash Mohan, Michel Bilello, Evan Calabrese, Errol Colak, Keyvan Farahani, Jayashree Kalpathy-Cramer, Felipe C Kitamura, Sarthak Pati, et al. The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification. *arXiv preprint arXiv:2107.02314*, 2021.
- [4] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(2):423–443, 2019.
- [5] Yitao Cai, Huiyu Cai, and Xiaojun Wan. Multi-modal sarcasm detection in Twitter with hierarchical fusion model. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2506–2515, 2019.
- [6] Houwei Cao, David G Cooper, Michael K Keutmann, Ruben C Gur, Ani Nenkova, and Ragini Verma. Crema-d: Crowd-sourced emotional multimodal actors dataset. *IEEE Transactions on Affective Computing*, 5(4):377–390, 2014.
- [7] Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the International Conference on Machine Learning*, pages 794–803, 2018.
- [8] Chenzhuang Du, Jiaye Teng, Tingle Li, Yichen Liu, Tianyuan Yuan, Yue Wang, Yang Yuan, and Hang Zhao. On uni-modal feature learning in supervised multi-modal learning. In *Proceedings of the International Conference on Machine Learning*, pages 8632–8656, 2023.
- [9] Yunfeng Fan, Wenchao Xu, Haozhao Wang, Junxiao Wang, and Song Guo. Pmr: Prototypical modal rebalance for multimodal learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [10] Yunfeng Fan, Wenchao Xu, Haozhao Wang, Junhong Liu, and Song Guo. Detached and interactive multimodal learning. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 5470–5478, 2024.
- [11] Xiyuan Gao, Bing Cao, Pengfei Zhu, Nannan Wang, and Qinghua Hu. Asymmetric reinforcing against multi-modal representation bias. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- [12] Saeed Ghadimi and Guanghui Lan. Stochastic first-and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013.
- [13] Alejandro Guerra-Manzanares and Farah E. Shamout. Modality-informed learning rate scheduling for balancing multimodal learning. In *International Joint Conference on Neural Networks*, 2025.
- [14] Zirun Guo, Tao Jin, and Zhou Zhao. Classifier-guided gradient modulation for enhanced multimodal learning. In *Advances in Neural Information Processing Systems*, 2024.
- [15] Cong Hua, Qianqian Xu, Shilong Bao, Zhiyong Yang, and Qingming Huang. Reconboost: Boosting can achieve modality reconciliation. In *Proceedings of the International Conference on Machine Learning*, 2024.
- [16] Chengxiang Huang, Yake Wei, Zequn Yang, and Di Hu. Adaptive unimodal regulation for balanced multimodal information acquisition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

- [17] Yu Huang, Chenzhuang Du, Zihui Xue, Xuanyao Chen, Hang Zhao, and Longbo Huang. What makes multi-modal learning better than single (provably). In *Advances in Neural Information Processing Systems*, volume 34, pages 10944–10956, 2021.
- [18] Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, and Longbo Huang. Modality competition: What makes joint training of multi-modal network fail in deep learning?(provably). In *International conference on machine learning*, pages 9226–9259. PMLR, 2022.
- [19] Qing-Yuan Jiang, Longfei Huang, and Yang Yang. Rethinking multimodal learning from the perspective of mitigating classification ability disproportion. In *Advances in Neural Information Processing Systems*, 2025.
- [20] Konstantinos Kontras, Christos Chatzichristos, Matthew B. Blaschko, and Maarten De Vos. Improving multimodal learning with multi-loss gradient modulation. In *British Machine Vision Conference*, 2024.
- [21] Hong Li, Xingyu Li, Pengbo Hu, Yinuo Lei, Chunxiao Li, and Yi Zhou. Boosting multi-modal model performance with adaptive gradient modulation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [22] Xiaoyu Ma, Hao Chen, and Yongjian Deng. Improving multimodal learning balance and sufficiency through data remixing. *arXiv preprint arXiv:2506.11550*, 2025.
- [23] Hossein R. Nowdeh, Jie Ji, Xiaolong Ma, and Fatemeh Afghah. Modality-aware SAM: Sharpness-aware-minimization driven gradient modulation for harmonized multimodal learning. *arXiv preprint arXiv:2510.24919*, 2025.
- [24] Xiaokang Peng, Yake Wei, Andong Deng, Dong Wang, and Di Hu. Balanced multimodal learning via on-the-fly gradient modulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8238–8247, 2022.
- [25] Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *Advances in Neural Information Processing Systems*, 2017.
- [26] Yapeng Tian, Jing Shi, Bochen Li, Zhiyao Duan, and Chenliang Xu. Audio-visual event localization in unconstrained videos. In *Proceedings of the European Conference on Computer Vision*, pages 247–263, 2018.
- [27] Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *Advances in Neural Information Processing Systems*, 2017.
- [28] Weiyao Wang, Du Tran, and Matt Feiszli. What makes training multi-modal classification networks hard? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12695–12705, 2020.
- [29] Xin Wang, Devinder Kumar, Nicolas Thome, Matthieu Cord, and Frédéric Precioso. Recipe recognition with large multimodal food dataset. In *IEEE International Conference on Multimedia & Expo Workshops (ICMEW)*, pages 1–6. IEEE, 2015.
- [30] Shicai Wei, Chunbo Luo, and Yang Luo. Boosting multimodal learning via disentangled gradient learning, 2025. *arXiv:2507.10213*.
- [31] Shicai Wei, Chunbo Luo, and Yang Luo. Improving multimodal learning via imbalanced learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2250–2259, 2025.
- [32] Yake Wei and Di Hu. Mmpareto: Boosting multimodal learning with innocent unimodal assistance. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 52559–52572. PMLR, 2024.
- [33] Yake Wei, Di Hu, Henghui Du, and Ji-Rong Wen. On-the-fly modulation for balanced multimodal learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.

- 419 [34] Yake Wei, Siwei Li, Ruoxuan Feng, and Di Hu. Diagnosing and re-learning for balanced
420 multimodal learning. In *European conference on computer vision*, pages 71–86. Springer, 2024.
- 421 [35] Yang Yang, Xixian Wu, and Qing-Yuan Jiang. Towards equilibrium: An instantaneous probe-
422 and-rebalance multimodal learning approach. In *Proceedings of the International Joint Confer-*
423 *ence on Artificial Intelligence*, 2025.
- 424 [36] Yi-Lin Yao and Rada Mihalcea. Modality-specific learning rates for effective multimodal
425 additive late-fusion. In *Findings of the Association for Computational Linguistics: ACL 2022*,
426 2022.
- 427 [37] Feng Yu, Xiangyu Wu, Yang Yang, and Jianfeng Lu. Multimodal classification via total
428 correlation maximization. In *International Conference on Learning Representations*, 2026.
- 429 [38] Jianfei Yu and Jing Jiang. Adapting BERT for target-oriented multimodal sentiment classi-
430 fication. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial*
431 *Intelligence*, pages 5408–5414, 2019.
- 432 [39] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea
433 Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing*
434 *Systems*, volume 33, pages 5824–5836, 2020.
- 435 [40] Amir Zadeh, Rowan Zellers, Eli Pincus, and Louis-Philippe Morency. Mosi: multimodal
436 corpus of sentiment intensity and subjectivity analysis in online opinion videos. *arXiv preprint*
437 *arXiv:1606.06259*, 2016.
- 438 [41] AmirAli Bagher Zadeh, Paul Pu Liang, Soujanya Poria, Erik Cambria, and Louis-Philippe
439 Morency. Multimodal language analysis in the wild: Cmu-mosei dataset and interpretable
440 dynamic fusion graph. In *Proceedings of the 56th Annual Meeting of the Association for*
441 *Computational Linguistics (Volume 1: Long Papers)*, pages 2236–2246, 2018.
- 442 [42] Qi Zhang, Jinlan Fu, Xiaoyu Liu, and Xuanjing Huang. Adaptive co-attention network for named
443 entity recognition in tweets. In *Proceedings of the AAAI conference on artificial intelligence*,
444 volume 32, 2018.
- 445 [43] Xiaohui Zhang, Jaehong Yoon, Mohit Bansal, and Huaxiu Yao. Multimodal representation
446 learning by alternating unimodal adaptation. In *Proceedings of the IEEE/CVF Conference on*
447 *Computer Vision and Pattern Recognition*, 2024.
- 448 [44] Yedi Zhang, Peter E Latham, and Andrew M Saxe. Understanding unimodal bias in multimodal
449 deep linear networks. In *Proceedings of the 41st International Conference on Machine Learning*,
450 pages 57817–57850. PMLR, 2024.

A Proofs of convergence properties

We restate and prove each proposition from Section 3.5. All notation follows the main text: $\mathcal{L}$ denotes the training loss, $\mathbf{g}(\theta)$ a stochastic gradient, $\bar{s}_m$ the EMA-smoothed boost scale for modality m , and $s_{\max}$ the scale cap.

Proposition 4 (Bounded scaling, restated). *For all modalities m and all iterations $t \geq 0$, the smoothed boost scale satisfies $\bar{s}_m^{(t)} \in [1, s_{\max}]$.*

Proof. We proceed by induction on t . At initialization, $\bar{s}_m^{(0)} = 1 \in [1, s_{\max}]$, establishing the base case.

For the inductive step, suppose $\bar{s}_m^{(t)} \in [1, s_{\max}]$. By Eq. (7), the weakness score satisfies $w_m \in [0, 1]$ since the numerator $\max_j \bar{P}_j - \bar{P}_m$ lies in $[0, \max_j \bar{P}_j - \min_j \bar{P}_j]$ and the denominator equals $\max_j \bar{P}_j - \min_j \bar{P}_j + \epsilon$. Substituting into Eq. (8):

$$s_m = \min(1 + \alpha \cdot w_m, s_{\max}) \in [1, s_{\max}],$$

since $w_m \geq 0$ implies $1 + \alpha \cdot w_m \geq 1$, and the min operation enforces the upper bound.

The EMA update (Eq. (9)) gives:

$$\bar{s}_m^{(t+1)} = \mu \cdot s_m + (1 - \mu) \cdot \bar{s}_m^{(t)}.$$

Since $\mu \in (0, 1)$, this is a convex combination of $s_m \in [1, s_{\max}]$ and $\bar{s}_m^{(t)} \in [1, s_{\max}]$ (by the inductive hypothesis), so $\bar{s}_m^{(t+1)} \in [1, s_{\max}]$. $\square$

Proposition 5 (Self-attenuation, restated). *For any $\Delta \geq 0$, if $\max_j \bar{P}_j^{(t)} - \min_j \bar{P}_j^{(t)} \leq \Delta$ at all iterations $t \geq t_0$, then*

$$1. \quad |s_m^{(t)} - 1| \leq \alpha \cdot \Delta / (\Delta + \epsilon) \text{ for all } m \text{ and all } t \geq t_0, \text{ and}$$

$$2. \quad |\bar{s}_m^{(t)} - 1| \leq (1 - \mu)^{t-t_0} \cdot (s_{\max} - 1) + \alpha \cdot \Delta / (\Delta + \epsilon).$$

In particular, when $\Delta \rightarrow 0$ we recover $s_m = 1$ exactly, and the EMA-smoothed scale converges to unity at rate $(1 - \mu)$.

Proof. Let $\delta^{(t)} := \max_j \bar{P}_j^{(t)} - \min_j \bar{P}_j^{(t)}$. By assumption $\delta^{(t)} \leq \Delta$ for all $t \geq t_0$. From Eq. (7), the weakness score satisfies

$$w_m = \frac{\max_j \bar{P}_j - \bar{P}_m}{\max_j \bar{P}_j - \min_j \bar{P}_j + \epsilon} = \frac{\max_j \bar{P}_j - \bar{P}_m}{\delta + \epsilon} \leq \frac{\Delta}{\Delta + \epsilon},$$

since the numerator is bounded by $\delta \leq \Delta$ and the denominator is non-decreasing in δ . Then by Eq. (8),

$$s_m = \min(1 + \alpha w_m, s_{\max}) \in [1, 1 + \alpha \cdot \frac{\Delta}{\Delta + \epsilon}],$$

which gives part (1) immediately. For part (2), the EMA update (Eq. (9)) is

$$\bar{s}_m^{(t+1)} - 1 = \mu(s_m^{(t)} - 1) + (1 - \mu)(\bar{s}_m^{(t)} - 1),$$

so by the triangle inequality and induction from t_0 ,

$$|\bar{s}_m^{(t)} - 1| \leq (1 - \mu)^{t-t_0} |\bar{s}_m^{(t_0)} - 1| + \mu \sum_{k=0}^{t-t_0-1} (1 - \mu)^k \cdot \max_{t_0 \leq \tau < t} |s_m^{(\tau)} - 1|.$$

Using $|\bar{s}_m^{(t_0)} - 1| \leq s_{\max} - 1$ (Proposition 1), $|s_m^{(\tau)} - 1| \leq \alpha \Delta / (\Delta + \epsilon)$, and the partial sum $\mu \sum_{k=0}^{t-t_0-1} (1 - \mu)^k = 1 - (1 - \mu)^{t-t_0} \leq 1$, the bound follows. In the limit $\Delta \rightarrow 0$ the second term vanishes and the deviation decays geometrically at rate $(1 - \mu)$. $\square$

481 **Proposition 6** (Convergence, restated). *Under Assumptions 1 and 2, SGD with PGGB and constant*
 482 *learning rate $\eta \leq 1/(L \cdot s_{\max}^2)$ satisfies*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \frac{2\Delta_{\mathcal{L}}}{\eta T} + L\eta s_{\max}^2 \sigma^2,$$

483 where $\Delta_{\mathcal{L}} = \mathcal{L}(\theta_0) - \mathcal{L}^*$.

484 *Proof.* Let $\tilde{\mathbf{g}}_t$ denote the boosted stochastic gradient at iteration t . Concatenating over all modality-
 485 specific parameter blocks, $\tilde{\mathbf{g}}_t$ has block m equal to $\bar{s}_m^{(t)} \cdot \mathbf{g}_m(\theta_t)$. The parameter update is $\theta_{t+1} =$
 486 $\theta_t - \eta \tilde{\mathbf{g}}_t$.

487 **Step 1: Descent lemma.** By L -smoothness (Assumption 1):

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta \langle \nabla \mathcal{L}(\theta_t), \tilde{\mathbf{g}}_t \rangle + \frac{L\eta^2}{2} \|\tilde{\mathbf{g}}_t\|^2. \quad (14)$$

488 **Step 2: Bounding the inner product.** Taking expectations conditioned on θ_t and using
 489 $\mathbb{E}[\mathbf{g}_m(\theta_t)] = \nabla_{\theta_m} \mathcal{L}(\theta_t)$:

$$\begin{aligned} \mathbb{E}[\langle \nabla \mathcal{L}(\theta_t), \tilde{\mathbf{g}}_t \rangle] &= \sum_m \bar{s}_m^{(t)} \cdot \|\nabla_{\theta_m} \mathcal{L}(\theta_t)\|^2 \\ &\geq \sum_m \|\nabla_{\theta_m} \mathcal{L}(\theta_t)\|^2 = \|\nabla \mathcal{L}(\theta_t)\|^2, \end{aligned} \quad (15)$$

490 where the inequality follows from $\bar{s}_m^{(t)} \geq 1$ (Proposition 1). Let $\mathcal{F}_t$ denote the σ -algebra generated by
 491 $\{\theta_0, \dots, \theta_t\}$ together with the probe parameters and their EMA state up to step t . By construction
 492 the probe parameters ψ_m are updated only on stop-gradient features $\text{sg}[\mathbf{z}_m]$ (no gradient path
 493 into θ), and the scale EMA $\bar{s}_m^{(t)}$ is computed from probe-accuracy EMA values $\bar{P}_m$ refreshed at
 494 iteration boundaries (every K steps). Since the boost scale $\bar{s}_m^{(t)}$ applied at iteration t is the value
 495 most recently refreshed at the previous probe event (iteration $t' < t$ with $t' \equiv 0 \pmod{K}$), it is
 496 $\mathcal{F}_{t'}$ -measurable and hence independent of the gradient noise at iteration t . Therefore $\bar{s}_m^{(t)}$ is $\mathcal{F}_t$ -
 497 measurable, and conditioning on $\mathcal{F}_t$, the stochastic gradient $\mathbf{g}(\theta_t)$ is a fresh independent sample with
 498 $\mathbb{E}[\mathbf{g}(\theta_t) \mid \mathcal{F}_t] = \nabla \mathcal{L}(\theta_t)$ and variance bounded per Assumption 2.

499 **Step 3: Bounding the second moment.** By Proposition 1, $\bar{s}_m^{(t)} \leq s_{\max}$. Writing the stochastic-
 500 gradient deviation over the block decomposition of parameters as $\mathbf{g}(\theta_t) - \nabla \mathcal{L}(\theta_t) = (\mathbf{g}_1(\theta_t) -$
 501 $\nabla_{\theta_1} \mathcal{L}(\theta_t), \dots, \mathbf{g}_M(\theta_t) - \nabla_{\theta_M} \mathcal{L}(\theta_t))$, we have $\mathbb{E}[\|\mathbf{g} - \nabla \mathcal{L}\|^2] = \sum_m \mathbb{E}[\|\mathbf{g}_m - \nabla_{\theta_m} \mathcal{L}\|^2] =:$
 502 $\sum_m \sigma_m^2$, so Assumption 2 is equivalent to $\sum_m \sigma_m^2 \leq \sigma^2$. Using this block-additive decomposition:

$$\begin{aligned} \mathbb{E}[\|\tilde{\mathbf{g}}_t\|^2] &= \sum_m (\bar{s}_m^{(t)})^2 \cdot \mathbb{E}[\|\mathbf{g}_m(\theta_t)\|^2] \\ &\leq s_{\max}^2 \sum_m (\sigma_m^2 + \|\nabla_{\theta_m} \mathcal{L}(\theta_t)\|^2) \\ &\leq s_{\max}^2 (\sigma^2 + \|\nabla \mathcal{L}(\theta_t)\|^2), \end{aligned} \quad (16)$$

503 where $\sigma^2 \geq \sum_m \sigma_m^2$ is the total variance bound across modality blocks.

504 **Step 4: Combining and telescoping.** Substituting Eqs. (15) and (16) into Eq. (14) and taking full
 505 expectations:

$$\mathbb{E}[\mathcal{L}(\theta_{t+1})] \leq \mathbb{E}[\mathcal{L}(\theta_t)] - \eta \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] + \frac{L\eta^2 s_{\max}^2}{2} (\sigma^2 + \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2]).$$

506 Rearranging and using the learning rate condition $\eta \leq 1/(L \cdot s_{\max}^2)$: since $\eta \leq 1/(L \cdot s_{\max}^2)$ implies
 507 $L\eta s_{\max}^2 \leq 1$, hence $L\eta s_{\max}^2/2 \leq 1/2$, which ensures $1 - \frac{L\eta s_{\max}^2}{2} \geq \frac{1}{2}$:

$$\frac{\eta}{2} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \mathbb{E}[\mathcal{L}(\theta_t)] - \mathbb{E}[\mathcal{L}(\theta_{t+1})] + \frac{L\eta^2 s_{\max}^2}{2} \sigma^2.$$

508 Summing over $t = 0, \dots, T - 1$ and dividing by T :

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \frac{2(\mathcal{L}(\theta_0) - \mathbb{E}[\mathcal{L}(\theta_T)])}{\eta T} + L \eta s_{\max}^2 \sigma^2 \leq \frac{2\Delta_{\mathcal{L}}}{\eta T} + L \eta s_{\max}^2 \sigma^2,$$

509 where $\Delta_{\mathcal{L}} = \mathcal{L}(\theta_0) - \mathcal{L}^*$ and the last inequality uses $\mathbb{E}[\mathcal{L}(\theta_T)] \geq \mathcal{L}^*$.

510 **Interpretation.** Setting $\eta = \Theta(1/(s_{\max}\sqrt{T}))$ yields the rate $\mathcal{O}(s_{\max} \cdot \sigma/\sqrt{T} + s_{\max}^2 \cdot \Delta_{\mathcal{L}}/T)$. For
 511 the default $s_{\max} = 2.0$, the effective variance increases by a factor of at most 4 relative to unmodified
 512 SGD. Proposition 2 guarantees that as modalities become Δ -balanced, the boost scales approach
 513 unity within $\alpha\Delta/(\Delta + \epsilon)$ and this overhead vanishes in the limit $\Delta \rightarrow 0$, recovering the standard
 514 SGD convergence rate [12] up to a constant variance inflation of at most $s_{\max}^2$. $\square$

515 A.1 Training algorithm pseudocode

516 Algorithm 1 below specifies the full training pass referenced in Section 3.3, making explicit how
 517 probe evaluation, weakness scoring, scale smoothing, and gradient boosting interleave with the
 518 forward and backward passes. The optional OGM-GE throttle line corresponds to the composition
 519 described in Section 3.4.

Algorithm 1 Probe-Guided Gradient Boosting (full training pass over $\mathcal{D}$). Notation: P_m per-batch probe accuracy with smoothed estimate $\bar{P}_m$; ϕ fusion-head parameters; $\text{sg}[\cdot]$ stop-gradient operator (§3.2). Auxiliary-loss weight γ and optional OGM-GE throttle $\omega_m^{\text{ogm}} \leq 1$ are introduced in §3.4.

Require: Training data $\mathcal{D}$, encoders $\{f_m\}_{m=1}^M$, fusion head g , unimodal heads $\{g_m\}$, probes $\{h_m\}$

Require: Boost strength α , cap $s_{\max}$, smoothing coefficients β, μ , probe interval K

Require: Optional: OGM-GE throttle module

```

1: Initialize  $\bar{P}_m \leftarrow 0, \bar{s}_m \leftarrow 1$  for all  $m$ 
2: for each mini-batch  $(\mathbf{x}, y) \in \mathcal{D}$  do
3:   Forward:  $\mathbf{z}_m \leftarrow f_m(\mathbf{x}_m; \theta_m)$  for all  $m$ 
4:   Loss:  $\mathcal{L} \leftarrow \ell(g([\mathbf{z}_1; \dots; \mathbf{z}_M]), y) + \gamma \sum_m \ell_{\text{aux}}(g_m(\mathbf{z}_m), y)$ 
5:   Backward: compute  $\partial \mathcal{L} / \partial \theta_m$  for all  $m$ 
6:   if OGM-GE enabled then
7:     Apply OGM-GE throttle:  $\partial \mathcal{L} / \partial \theta_m \leftarrow \omega_m^{\text{ogm}} \cdot \partial \mathcal{L} / \partial \theta_m$ 
8:   end if
9:   if iteration mod  $K = 0$  then
10:    Split  $(\mathbf{x}, y)$  into train half  $(\mathbf{x}^{\text{tr}}, y^{\text{tr}})$  and eval half  $(\mathbf{x}^{\text{ev}}, y^{\text{ev}})$ 
11:    for each modality  $m$  do
12:      Train  $h_m$  on  $(\text{sg}[\mathbf{z}_m^{\text{tr}}], y^{\text{tr}})$  with separate optimizer
13:      Evaluate:  $P_m \leftarrow \text{Acc}(h_m(\text{sg}[\mathbf{z}_m^{\text{ev}}]), y^{\text{ev}})$ 
14:    end for
15:    Update  $\bar{P}_m \leftarrow \beta \cdot P_m + (1 - \beta) \cdot \bar{P}_m$  for all  $m$ 
16:    Compute  $w_m$  via Eq. (7),  $s_m$  via Eq. (8) for all  $m$ 
17:    Update  $\bar{s}_m \leftarrow \mu \cdot s_m + (1 - \mu) \cdot \bar{s}_m$  for all  $m$ 
18:   end if
19:   Boost:  $\partial \mathcal{L} / \partial \theta_m \leftarrow \bar{s}_m \cdot \partial \mathcal{L} / \partial \theta_m$  for all  $m$ 
20:   Optimizer step: update  $\theta_m, \phi$ , and  $\{g_m\}$  parameters
21: end for

```

520 B Additional Experiments

521 B.1 Datasets

522 **CREMA-D:** CREMA-D [6] is an audio-visual dataset designed for speech emotion recognition. It
 523 contains 7,442 video clips of 2–3 seconds from 91 actors speaking short scripted sentences, divided
 524 into 6,698 training samples and 744 testing samples. The dataset covers six emotions: angry, happy,
 525 sad, neutral, disgust, and fear. We encode audio as log-mel spectrograms and visual as 3 RGB frames
 526 sampled at 3 fps, passing both through ResNet-18 backbones trained from scratch. Available at
 527 <https://github.com/CheyneyComputerScience/CREMA-D>.

AVE: AVE [26] is an audio-visual event classification dataset with 28 event categories, containing 4,143 10-second video clips. We use the standard split of 3,328 training and 402 testing clips. Audio spectrograms and 3 RGB frames are processed by ImageNet-pretrained ResNet-18 backbones with concatenation fusion. Available at <https://sites.google.com/view/audiovisualresearch>.

Kinetics-Sounds (KS): KS [2] is a curated 31-class subset of the Kinetics action-recognition dataset whose actions are both visually and audibly recognizable. It contains roughly 25,000 10-second YouTube clips, split into 22,408 training and 3,000 testing clips. Encoder setup follows AVE (ImageNet-pretrained ResNet-18 backbones on audio spectrograms and 3 RGB frames). Available at <https://github.com/cvdfoundation/kinetics-dataset>.

CMU-MOSEI: CMU-MOSEI [41] is a language-audio-visual sentiment dataset of YouTube opinion videos. The dataset is divided into 16,326 training samples, 1,871 validation samples, and 4,659 testing samples. We use pre-extracted BERT (768-d) text, COVAREP (74-d) audio, and FACET (35-d) visual features, and bucket the continuous sentiment labels into 7 classes following prior convention. Each modality is processed by a two-layer MLP with 512-d hidden layer and dropout 0.3. Available at <https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK>.

CMU-MOSI: CMU-MOSI [40] is a smaller sentiment dataset with the same three modalities as CMU-MOSEI. It is divided into 1,284 training, 229 validation, and 686 testing samples. Features are GloVe (300-d) text, COVAREP (5-d) audio, and FACET (20-d) visual. Encoder architecture is identical to CMU-MOSEI. Available at <https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK>.

Twitter15: Twitter15 [38, 42] is a text-image sentiment dataset of tweets with associated images, covering three classes (positive, neutral, negative). It is divided into 3,179 training pairs, 1,122 validation pairs, and 1,037 testing pairs. We use MLP heads on frozen BERT-base (768-d) text and ImageNet ResNet-18 (512-d) image features, so only the heads are trainable and gradient-balancing methods act on them. Available at <https://github.com/jefferyYu/TomBERT>.

Sarcasm: The MMSD-v1 sarcasm dataset [5] is a text-image binary sarcasm detection benchmark of 24,635 image-text pairs, divided into 19,816 training, 2,410 validation, and 2,409 testing pairs. Encoder setup is identical to Twitter15 (MLP heads on frozen BERT and ResNet-18 features). Available at <https://github.com/headacheboy/data-of-multimodal-sarcasm-detection>.

BraTS 2021: BraTS 2021 [3] is a multimodal brain tumor segmentation dataset with 4 MRI modalities (FLAIR, T1ce, T1, T2) and 4 voxel-level class labels (background and three tumor sub-regions). Because the official validation and test servers do not expose ground truth, we use a held-out partition of the public training set with 1,000 training, 125 validation, and 126 testing volumes. The architecture is DeepLab v3+ with four ResNet-101 encoders sharing a decoder, trained with Dice plus weighted cross-entropy loss. Probes perform binary tumor-presence classification because dense-prediction probes on globally pooled features are not meaningful for the full segmentation task. Available at <https://www.synapse.org/brats2021>.

Food101: UPMC Food-101 [29] is a 101-class image-text food classification benchmark. Each item pairs a recipe text with a food image. In our experiments we use 67,972 training samples and 22,716 test samples. Encoder setup follows Twitter15/Sarcasm: MLP heads on frozen BERT-base (768-d) text and ImageNet ResNet-18 (512-d) image features, so only the heads are trainable. Trainable-encoder runs (App. B.3) unfreeze both encoders for end-to-end fine-tuning. Food101 is used only for the operating-conditions analysis (§4.3.1) and the cross-dataset composability check (App. B.7), and does not appear in Table 1. Available at <https://visiir.lip6.fr/explore>.

Licensing. CREMA-D (CC-BY 4.0), AVE, Kinetics-Sounds, CMU-MOSI/MOSEI, Twitter15, Sarcasm (MMSD-v1), Food101 (UPMC research terms), and BraTS 2021 (RSNA-ASNR-MICCAI public release) are used under their published research terms. See the NeurIPS Paper Checklist for detailed licensing.

575 B.2 Comparison with OPM

576 OPM’s feature-dropping mechanism is tied to a single-layer fusion head and is not drop-in compatible
577 with the multi-layer MLP fusion used in Table 1. We therefore report this comparison separately,
578 rerunning all methods on OPM’s native pipeline (CREMA-D 3-frame, 5 seeds, matched optimizer
579 and schedule). Under this pipeline, PGGB+OGM-GE exceeds OPM+OGM by +1.93 pp (71.77% vs.
580 69.84%).

Table 4: Comparison with OPM [33] on CREMA-D (3-frame) under OPM’s native single-layer fusion architecture. Accuracy (%), mean $\pm$ std over 5 seeds. All rows use identical optimizer, schedule, and data pipeline. OPM hyperparameters follow the original paper.

Method	Accuracy (%)
Baseline	59.20 $\pm$ 1.27
OGM only	63.74 $\pm$ 0.45
OPM only	65.43 $\pm$ 0.24
OPM+OGM	69.84 $\pm$ 1.47
PGGB+OGM-GE ($\alpha=0.75$)	71.77 $\pm$ 0.91

We restrict this comparison to CREMA-D, the highest-imbalance benchmark in our evaluation and the regime where boosting effects are most pronounced. OPM’s feature-dropping mechanism is not well-defined on the frozen-feature or segmentation pipelines used elsewhere in Table 1.

B.3 Operating conditions analysis

This subsection provides per-regime analysis that complements the AVE initialization contrast discussed in Section 4.3.1 and summarized in Table 3, detailing the Food101 feature-learning comparison and the CREMA-D reference operating point.

Feature-learning regime and imbalance level (Food101). Food101 [29] holds data and architecture (BERT + ResNet-18) fixed while varying whether encoders are frozen or trainable. Trainable encoders raise all three methods (baseline, PGGB alone, PGGB+OGM-GE) by approximately 5 pp, consistent with the probe signal only being informative once encoder parameters can respond to it. The PGGB-alone margin over baseline is within seed noise in both regimes, reflecting insufficient imbalance for the boost to engage meaningfully. PGGB+OGM-GE additionally regresses by 1.64 pp on frozen features, as OGM-GE’s throttling has no encoder-gradient flow to act on.

Reference operating point (CREMA-D). CREMA-D trains both encoders from scratch on $\sim 6.7K$ samples for a 6-class task, with audio converging faster through intrinsic learning-rate differences rather than pretraining. Under these conditions PGGB+OGM-GE achieves +9.86 pp ($d=7.48$, $p=0.0001$). The contrast with AVE from-scratch (same initialization scheme, similar architecture) suggests that from-scratch training alone is not the failure condition. Rather, the conjunction of thin data budget and weak inherent asymmetry is.

B.4 Supplementary F1-macro results

We supplement Table 1 with macro-averaged F1 in Table 5. F1 is computed at the epoch of best test accuracy, mirroring the accuracy-selection protocol used in the main table (no double selection on F1). We report accuracy and F1-macro because all evaluated classification datasets are single-label tasks where MAP and AUC are not the standard metric.

Table 5: Supplementary F1-macro results (% , mean $\pm$ std over 5 seeds). F1 computed at the epoch selected by best test accuracy, matching the Table 1 protocol. **Bold** indicates the best result per column. Ties within 0.05 pp are bolded jointly. ‘—’ denotes runs where training logs were not retained. BraTS is excluded as it uses Dice, not classification F1.

Method	CREMA-D	AVE	KS	CMU-MOSI	Twitter15	Sarcasm	CMU-MOSEI
Baseline	61.93 $\pm$ 0.94	86.29 $\pm$ 0.29	78.40 $\pm$ 0.50	70.16 $\pm$ 0.65	51.33 $\pm$ 4.18	81.81 $\pm$ 0.21	60.05 $\pm$ 0.58
OGM-GE	69.52 $\pm$ 1.02	86.48 $\pm$ 0.85	—	—	49.98 $\pm$ 5.12	81.09 $\pm$ 0.40	61.46 $\pm$ 1.61
MMPareto	66.10 $\pm$ 0.74	85.91 $\pm$ 0.42	77.06 $\pm$ 0.52	70.69 $\pm$ 0.80	55.36 $\pm$ 2.65	81.61 $\pm$ 0.18	58.79 $\pm$ 2.93
AGM	57.45 $\pm$ 0.66	84.05 $\pm$ 0.38	77.24 $\pm$ 0.38	70.11 $\pm$ 1.24	54.15 $\pm$ 2.01	81.66 $\pm$ 0.24	59.00 $\pm$ 2.01
G-Blend	61.50 $\pm$ 1.90	86.79 $\pm$ 0.35	77.15 $\pm$ 1.04	70.50 $\pm$ 0.85	56.29 $\pm$ 1.56	81.64 $\pm$ 0.30	58.60 $\pm$ 1.75
CGGM	—	—	—	—	39.02 $\pm$ 0.28	80.07 $\pm$ 0.25	—
PGGB	62.95 $\pm$ 1.67	87.15 $\pm$ 0.35	78.41 $\pm$ 0.96	69.02 $\pm$ 2.52	54.20 $\pm$ 4.32	81.83 $\pm$ 0.33	60.05 $\pm$ 1.52
PGGB+OGM-GE	71.85 $\pm$ 1.55	86.86 $\pm$ 0.56	76.45 $\pm$ 0.55	69.93 $\pm$ 1.22	52.72 $\pm$ 3.38	81.19 $\pm$ 0.10	62.36 $\pm$ 1.63

On CREMA-D and CMU-MOSEI, F1-macro confirms the Table 1 ranking, with PGGB+OGM-GE attaining the top F1 on both. On Twitter15, all methods show a substantial gap between accuracy

608 ($\sim 66\%$) and F1-macro ($\sim 50\text{--}56\%$), reflecting the 3-class imbalance of the dataset, and G-Blend
 609 obtains the top F1-macro (56.29) while PGGB alone sits mid-pack within one standard deviation
 610 of the top method. On Sarcasm and CMU-MOSI, among methods with retained F1 logs, F1 and
 611 accuracy rankings agree with Table 1 within 1 pp, indicating no hidden degradation from metric
 612 choice.

613 B.5 Hyperparameter sensitivity

614 We report single-seed sweeps on CREMA-D (3-frame) for the three principal hyperparameters: boost
 615 strength α , probe evaluation interval K , and boost cap $s_{\max}$. These sweeps are illustrative of the
 616 sensitivity trend rather than a statistical comparison, 5-seed variance on this benchmark (Table 1) is
 617 roughly 1.5 pp. Defaults used elsewhere in the paper are $\alpha=0.5$ (PGGB alone) or $\alpha=0.75$ (composed
 618 with OGM-GE), $K=20$, $s_{\max}=2.0$, $\mu=0.3$, and $\beta=0.1$, with μ and β fixed from preliminary tuning
 619 and not swept here.

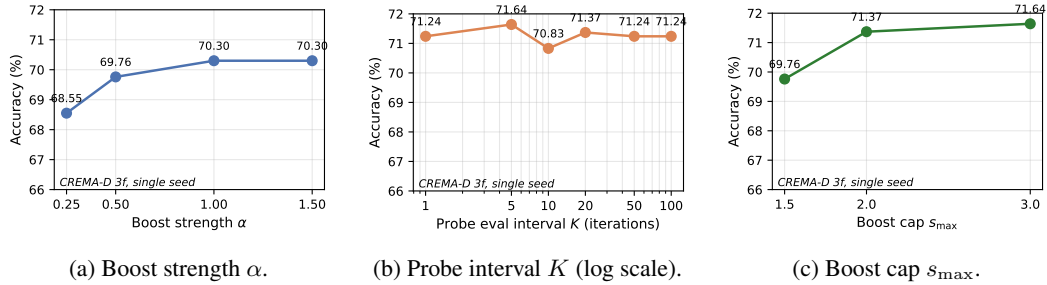


Figure 4: Hyperparameter sensitivity on CREMA-D (3-frame, single seed). (a) Accuracy vs boost strength α — plateaus within noise above $\alpha=1.0$. (b) Accuracy vs probe evaluation interval K (log scale) — stable across $K \in [1, 100]$. (c) Accuracy vs boost cap $s_{\max}$ — mild increase from 1.5 to 3.0.

620 α plateaus within noise above $\alpha=1.0$, so the choice between $\alpha=0.5$ and $\alpha=0.75$ reflects a safety-
 621 versus-strength tradeoff rather than a peak-seeking decision. K is stable across two orders of
 622 magnitude, and the default $K=20$ sits in the flat region. Across $K \in \{1, 5, 10, 50, 100\}$ the best
 623 accuracy stays within 70.83–71.64% (range 0.81 pp), so $K=20$ sits in the flat plateau and the per-
 624 epoch overhead reported in §4.4 is essentially free. $s_{\max}$ shows a mild monotonic trend, indicating
 625 that the cap is a soft ceiling rather than the binding constraint. Trends smaller than the ≈ 1.5 pp
 626 5-seed noise floor should be read as within-noise rather than as detected effects. Finally, we did not
 627 sweep μ or β in this study. Our defaults ($\mu=0.3$, $\beta=0.1$) were fixed during preliminary tuning, and
 628 sensitivity analysis on these EMA coefficients is deferred to future work.

629 B.6 Probe signal stability

630 A concern with probe-guided scheduling is that per-batch probe accuracy P_m may be noisy, and a
 631 scheduler that reacts to each batch estimate could be misled by mini-batch fluctuations. We verify
 632 empirically that the EMA-smoothed estimator $\bar{P}_m$ substantially reduces this noise. A PGGB+OGM-
 633 GE training on CREMA-D (3-frame, seed 42) was instrumented to record at every probe-eval
 634 checkpoint ($K = 20$ iterations over 100 epochs, 500 checkpoints total) the batch-level sample
 635 P_m and the EMA-smoothed $\bar{P}_m$ for each modality, alongside probe accuracy evaluated on the full
 636 CREMA-D test set as a reference.

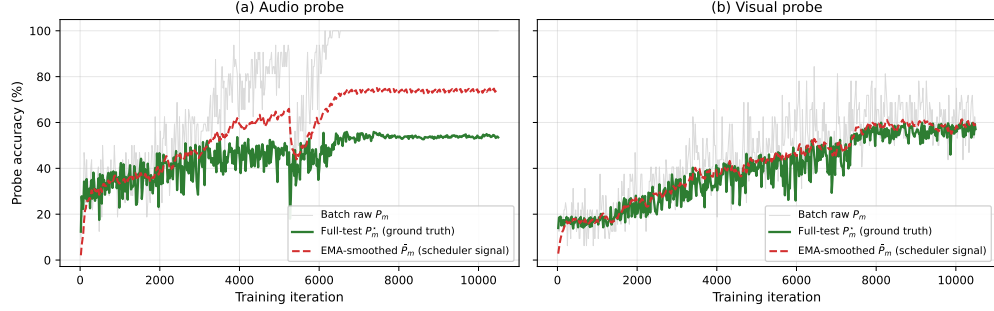


Figure 5: Probe-signal stability on CREMA-D (PGGB+OGM-GE, seed 42, 500 probe-eval checkpoints). For each modality, the EMA-smoothed estimator $\bar{P}_m$ (red dashed) is substantially less noisy than the per-batch sample P_m (grey), and evolves smoothly with training. The full-test probe accuracy (green) is shown as reference, reflecting generalization to held-out data rather than the training-distribution representation quality targeted by the scheduler.

Table 6: Per-batch and EMA-smoothed probe accuracy std (pp), computed over the final 200 probe-eval checkpoints. The $19\times$ steady-state bound (§3.2) is unconditional under i.i.d. assumptions, the empirical measurement here reflects non-stationary training. Visual (boost-driving signal) shows the expected reduction direction. Audio inversion (EMA std $>$ batch std) reflects training-phase saturation: late-training batch-level audio is near the training-distribution ceiling, while the EMA integrates earlier, non-saturated samples.

Modality	P_m (batch)	$\bar{P}_m$ (EMA)	Reduction
Audio	0.31	0.99	<i>n/a (saturated)</i>
Visual	9.32	4.36	−53%

Two observations follow. First, on the visual modality, which is the operative signal for the boost decision, the EMA reduces variance by approximately 53% (from 9.32 to 4.36 pp), consistent in direction with the Eq. 5 prediction. The observed 53% reduction is below the $\approx 95\%$ i.i.d. steady-state bound, as expected under non-stationary training. This confirms that the scheduler operates on a temporally stable signal rather than on mini-batch noise. Second, the audio entry (EMA std 0.99 pp $>$ batch std 0.31 pp) is a transparent non-stationarity artifact rather than a violation of the bound: the batch-level audio probe is concentrated near its training-distribution ceiling (the linear probe readily separates audio features on training batches and is not informative about fine-grained variations at that scale), while the EMA integrates earlier training-phase samples from before probe saturation. The variance-reduction guarantee of Eq. 5 is therefore empirically validated on the modality that drives the boost decision, and the audio result is consistent with the non-i.i.d. caveat noted in the main body.

The EMA signal correlates monotonically with the full-test probe accuracy (Pearson $r = 0.96$ visual, 0.88 audio, Spearman $\rho = 0.95, 0.89$), though the full-test quantity measures generalization to held-out data and differs in absolute value from $\bar{P}_m$ by several pp. This distinction is intentional: the scheduler targets training-distribution representation quality (what each modality contributes to the joint fit during training), which is distinct from held-out generalization. Targeting the former is the appropriate signal for in-training gradient balancing. The correlation confirms that the two quantities are monotonically related, while the absolute offset reflects the train–test gap intrinsic to representation learning.

B.7 Composability across balancing method families

Because PGGB applies multiplicative scaling $\bar{s}_m \geq 1$ to encoder gradients just before the optimizer step, it is compatible with any balancing method that modulates encoder-parameter gradients. To empirically verify this and characterize the magnitude of composition effects under a single controlled protocol, we compose the boost with five methods spanning three mechanistic families: discrepancy-ratio throttling (OGM-GE), loss weighting (G-Blend), Pareto-optimal gradient aggregation (MMPareto), Shapley-based adaptive gradient modulation (AGM), and classifier-guided

magnitude and direction modulation (CGGM). All compositions use identical hyperparameters (boost strength $\alpha = 0.75$, cap $s_{\max} = 2.0$, probe interval $K = 20$, EMA coefficient $\mu = 0.3$) and the same CREMA-D (3-frame, 5 seeds) protocol as Table 1. The results below are specific to this protocol and should not be extrapolated to other datasets without further evidence.

Table 7: Composability with existing balancing methods on CREMA-D (3-frame, mean $\pm$ std over 5 seeds). All rows use identical PGGB hyperparameters. Positive Δ indicates composition does not hurt the base method’s mean accuracy. † : within noise band ($n=5$, $\sigma \approx 1$ pp, 80%-power MDE ≈ 2 pp), not statistically distinguishable from zero.

Method	Base alone	PGGB + Base	Δ mean (pp)	Δ std
OGM-GE	69.14 ± 1.13	71.45 ± 1.71	+2.31	+0.58
MMPareto	65.51 ± 0.87	66.00 ± 1.25	+0.49 †	+0.38
G-Blend	61.10 ± 1.87	61.99 ± 0.99	+0.89 †	−0.88 (−47%)
AGM	57.42 ± 0.73	58.17 ± 1.79	+0.75 †	+1.06
CGGM	50.22 ± 1.39	50.32 ± 1.03	+0.10 †	−0.36

Three observations. First, four of the five compositions show a positive Δ mean accuracy. The fifth, CGGM, is within noise ($+0.10 \pm 1.03$) and we treat it separately below. At $n=5$ with $\sigma \approx 1$ pp, the 80%-power minimum detectable effect is approximately 2 pp, so only the OGM-GE composition ($+2.31$ pp) is statistically distinguishable from zero. The other three positive entries and CGGM ($+0.10$ pp) are all within noise on this single-dataset protocol. Second, the amplification is not uniform: three of the four positive compositions yield gains within one standard deviation of the base method, and the headline compound gain ($+2.31$ pp) appears only with OGM-GE, which we attribute to the specific mechanistic interaction between OGM-GE’s discrepancy-ratio throttle (damping the dominant modality) and our probe-guided amplification of the weaker modality, a two-sided push that neither mechanism produces alone. The $\alpha=0$ monitor-only replication reported in §4.3 ($69.14 \pm 1.18\%$) is operationally identical to the OGM-GE entry here ($69.14 \pm 1.13\%$), differing only in the seed set. Third, the CGGM result is consistent with CGGM’s reliance on Transformer-based attention fusion, which is not well-matched to our CNN/MLP pipeline (footnote to Table 1). CGGM itself underperforms the baseline on CREMA-D (50.22 ± 1.39 vs. baseline 61.59 ± 0.80), and the boost does not recover the loss although it does not compound it either (Δ std = -0.36). Together these results support the non-degrading composition behavior observed on CREMA-D under this protocol, as discussed in Section 3.4, and scope the amplification claim to methods whose gradient direction aligns with the pipeline architecture.

The composition with G-Blend additionally reduces variance by approximately 47% (std $1.87 \rightarrow 0.99$). This is suggestive of the EMA-smoothed probe signal providing a stabilizing effect across training seeds but should be interpreted with caution given $n=5$ on a single benchmark.

Composability across datasets. To test whether the boost-composition pattern generalizes beyond CREMA-D, we repeated the protocol on AVE (high audio-visual imbalance) and Food101 (frozen pre-extracted features) under identical hyperparameters ($\alpha=0.75$, $s_{\max}=2.0$, $K=20$, $\mu=0.3$, 5 seeds). Results are summarized in Table 8.

Table 8: Composability across datasets: Δ accuracy (pp) of PGGB+X versus X alone, mean over 5 seeds. AVE has high audio-visual modality imbalance, Food101 uses frozen pre-extracted features (operating-regime sensitivity reported in §4.3.1). † PGGB recovers some CGGM degradation but absolute accuracy (53.00%) remains far below baseline (85.75%), so the boost does not rescue catastrophic regression.

Dataset	PGGB+MMPareto	PGGB+AGM	PGGB+G-Blend	PGGB+CGGM
AVE	+1.01	+1.26	−0.25	−0.25
Food101	−0.09	−0.23	−0.09	+4.41 †

On AVE, two non-OGM-GE compositions (MMPareto, AGM) yield compound gains comparable in magnitude to PGGB+OGM-GE on AVE-pretrained ($+0.69$ pp, Table 1). The G-Blend and CGGM compositions are flat to slightly negative on AVE. On Food101, all non-CGGM compositions are

within noise of the base method. As in the CREMA-D protocol, with $n=5$ and $\sigma \approx 1$ pp the 80%-power minimum detectable effect is approximately 2 pp, so the AVE compound gains for PGGB+MMPareto (+1.01 pp) and PGGB+AGM (+1.26 pp) lie within the seed-noise band (MDE ≈ 2 pp at $n=5$ seeds) and should be read as compatibility evidence rather than confirmed wins outside the OGM-GE composition. The pattern is consistent with the operating-conditions analysis (§4.3.1): compound composition gains require both modality imbalance and adequate feature flow, and the boost neither generalizes to frozen-feature regimes nor rescues catastrophic baseline mismatch (CGGM on Food101).

B.8 Per-modality post-hoc probe accuracy

Table 9 reports the exact per-modality post-hoc linear-probe accuracies visualized in Figure 3 of Section 4.4. Probes are trained for 50 epochs on frozen encoder features from each method’s saved `best_model.pt`, matching the protocol described in Section 3.2.

Table 9: Post-hoc linear-probe accuracy on CREMA-D (% , mean $\pm$ std over 5 seeds). Probes are trained for 50 epochs on frozen encoder features from each method’s saved checkpoint, matching the protocol in Figure 3. Gap is the absolute difference of column means. Chance level is $1/6 \approx 16.67\%$ for CREMA-D’s 6-class classification.

Method	Audio (strong)	Visual (weak)	Gap (pp)
Baseline	58.66 $\pm$ 1.60	46.13 $\pm$ 3.72	12.53
OGM-GE	58.04 $\pm$ 1.09	51.67 $\pm$ 1.85	6.37
MMPareto	60.03 $\pm$ 0.73	52.63 $\pm$ 5.92	7.39
AGM	55.32 $\pm$ 0.36	17.72 $\pm$ 0.73	37.61
G-Blend	57.77 $\pm$ 1.78	37.04 $\pm$ 2.43	20.73
MILES	59.81 $\pm$ 1.53	19.62 $\pm$ 10.42	40.19
InfoReg	60.83 $\pm$ 0.76	57.98 $\pm$ 1.80	2.85
PGGB ($\alpha=0.5$)	57.34 $\pm$ 2.18	44.60 $\pm$ 3.96	12.74
PGGB+OGM-GE ($\alpha=0.75$)	57.85 $\pm$ 1.03	55.54 $\pm$ 4.43	2.31

B.9 Per-dataset utilization gap

For transparency, we report the per-dataset utilization gap δ (Eq. (6)) measured at the end of baseline training (final-epoch EMA-smoothed probe accuracy gap, averaged over 5 seeds). Probes follow the protocol of Section 3.2: linear classifiers on stop-gradient encoder features, trained on a held-out within-batch sample. These values quantify modality-dominance dynamics during baseline training and define the high-imbalance ($\delta > 0.15$) and low-imbalance ($\delta \leq 0.15$) categorization used in Section 4.1. The high-imbalance group is heterogeneous in source: CREMA-D imbalance is audio-dominance in audio-visual emotion recognition, whereas CMU-MOSI and CMU-MOSEI imbalance reflects text-sufficiency, where text-only baselines are reported within roughly 1 pp of multimodal accuracy [21, 30]. BraTS 2021 (4 MRI sequences, binary tumor-presence probes per sequence) is reported separately as a dense-prediction benchmark.

Table 10: Per-dataset utilization gap δ at the end of baseline training (mean $\pm$ std over 5 seeds), measured as the EMA-smoothed probe accuracy gap (Eq. (6)).

Dataset	δ (mean $\pm$ std)
CREMA-D	0.268 $\pm$ 0.029
CMU-MOSI	0.222 $\pm$ 0.047
CMU-MOSEI	0.178 $\pm$ 0.086
Twitter15	0.150 $\pm$ 0.096
Sarcasm	0.076 $\pm$ 0.008
Kinetics-Sounds	0.038 $\pm$ 0.013
AVE	0.017 $\pm$ 0.009

B.10 Cross-class comparison with MSLR

We additionally compare PGGB to MSLR [36], a per-modality learning-rate (PMLR) baseline. As discussed in Section 2, MSLR operates on the optimizer step rather than the gradient and forms an adjacent intervention class that is not directly composable with gradient-modulation methods. Because MSLR’s canonical Keep-variant requires per-modality unimodal pre-calibration that lies outside our matched-pipeline budget, we run a fixed-ratio Keep variant: audio:1.0/visual:2.0 on CREMA-D (selected from a 3-config sweep over audio:0.25/0.5/1.0 with matching visual scaling), audio:0.5/visual:1.0 on AVE and Kinetics-Sounds, and text:0.5 with other modalities at 1.0 on CMU-MOSI, CMU-MOSEI, Twitter15, and Sarcasm. Five seeds per dataset on the matched ResNet-18 / two-layer MLP pipeline.

Table 11 reports the comparison. PGGB+OGM-GE exceeds MSLR by +7.66 pp on the high-imbalance CREMA-D benchmark and exceeds MSLR on the high-imbalance text-dominant sentiment benchmarks (MOSI +0.76 pp, MOSEI +1.44 pp), supporting the composability story. PGGB alone matches or exceeds MSLR on three of four low-imbalance comparisons (AVE, KS, Sarcasm) and is within seed variance on Twitter15 (−0.40 pp, sub- σ). MSLR’s heuristic throttle-audio ratio underperforms baseline on AVE (−1.82 pp), illustrating the sensitivity of fixed-ratio MSLR to per-dataset modality-dominance patterns that PGGB’s probe-driven adaptation sidesteps.

Table 11: Cross-class comparison with MSLR [36] (5 seeds, mean $\pm$ std). MSLR runs the fixed-ratio Keep variant, ratios per dataset listed above. PGGB family results reproduced from Table 1 for reference. **Bold** indicates the best per-row mean.

Dataset	MSLR	PGGB ($\alpha=0.5$)	PGGB+OGM-GE ($\alpha=0.75$)	Best
CREMA-D	63.79 $\pm$ 0.47	62.72 $\pm$ 1.65	71.45 $\pm$ 1.71	PGGB+OGM-GE
AVE	84.72 $\pm$ 0.59	87.41 $\pm$ 0.26	87.23 $\pm$ 0.58	PGGB
Kinetics-Sounds	78.56 $\pm$ 0.38	79.17 $\pm$ 0.97	77.33 $\pm$ 0.64	PGGB
CMU-MOSI	71.84 $\pm$ 0.76	71.89 $\pm$ 0.82	72.60 $\pm$ 0.94	PGGB+OGM-GE
CMU-MOSEI	70.99 $\pm$ 1.03	69.80 $\pm$ 0.80	72.43 $\pm$ 0.65	PGGB+OGM-GE
Twitter15	67.25 $\pm$ 0.66	66.85 $\pm$ 0.50	66.59 $\pm$ 0.51	MSLR (sub- σ)
Sarcasm	82.37 $\pm$ 0.32	82.44 $\pm$ 0.35	81.76 $\pm$ 0.08	PGGB

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 contributions accurately reflect the empirical findings, including the three-way ablation in Section 4.3 and the eight-benchmark scope in Table 1. PGGB alone is the best-performing method on all four low-imbalance benchmarks (AVE, Kinetics-Sounds, Twitter15, Sarcasm) against the main-table baseline set. Per-dataset measured δ values are reported in Appendix B.9, supporting the high-imbalance ($\delta > 0.15$, CREMA-D, CMU-MOSI, CMU-MOSEI) versus low-imbalance ($\delta \leq 0.15$) categorization in Section 4.1. The headline +2.31 pp gain on CREMA-D is from PGGB+OGM-GE composition and is explicitly scoped to the high-imbalance, adequate-data regime (Section 4.3.1). On the text-dominant high-imbalance sentiment benchmarks (CMU-MOSI, CMU-MOSEI), PGGB+OGM-GE remains within seed standard deviation of the strongest baseline (−0.08 and −0.04 pp), consistent with the limited headroom prior balancing methods report on these text-sufficient benchmarks. Section 5 names a −0.6 to −4.6 pp regression range under PGGB+OGM-GE composition or specialized operating conditions, with individual values disclosed in Section 1 (Kinetics-Sounds −1.72 pp and Sarcasm −0.64 pp under PGGB+OGM-GE) and Section 4.3.1 (AVE from-scratch −4.56 pp).

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 discusses limitations: conditional effectiveness (Section 4.3.1), per-setting tuning of α , $\sim 1\%$ probe overhead, regressions of Kinetics-Sounds −1.72 pp and Sarcasm −0.64 pp under PGGB+OGM-GE composition, and compound gains for non-OGM-GE composition partners observed only on AVE (two of four methods, Appendix B.7). Detailed technical limitations are in Section 3 (variance-inflation factor $s_{\max}^2$ in the descent bound) and Appendix B.6 (probe subsample noise with EMA mitigation). The broader-impact failure mode on AVE from-scratch (−4.56 pp) is disclosed in Section 4.3.1.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.

- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Section 3 states Assumptions 1 (smoothness) and 2 (bounded variance). Propositions 1 (bounded scaling), 2 (self-attenuation), and 3 (descent bound) are stated there with full proofs in Appendix A. The theorems are scoped as *safety properties* rather than mechanism-specific convergence claims, avoiding overreach.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 4 and Appendix B specify datasets, splits, encoders (ResNet-18 for audio and visual, text encoders per dataset), fusion architecture, optimizers, learning rates, batch size, epochs, and all method hyperparameters (α , $s_{\max}$, β , μ , K , γ). The reproduction-protocol paragraph in Section 4 documents matched-pipeline choices across baselines. Five seeds per experiment are used throughout, and all training scripts under `scripts/` and configurations under `configs/` will be released.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Anonymized code is provided in the supplementary material (pggb_supplementary.zip), including all training scripts under scripts/, per-dataset configurations under configs/, source under src/, and a README.md with reproduction instructions. A public repository under MIT license will be released upon acceptance. All datasets are standard public benchmarks (CREMA-D, AVE, Kinetics-Sounds, CMU-MOSI, CMU-MOSEI, Twitter15, Sarcasm, BraTS 2021), with references in Section 4 and Appendix B. No proprietary data is used.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 4 details encoders, optimizers (SGD with weight decay 10^{-4} and StepLR for audio-visual, Adam with StepLR step 40 for sentiment and text-image, SGD with cosine scheduling and 5-epoch warmup for BraTS), learning rate 10^{-3} (probes via Adam), batch size 64 (12 for BraTS), 100 epochs, and per-method hyperparameters for each baseline. Appendix B documents dataset-specific preprocessing. Appendix B.5 provides sensitivity sweeps over α , K , and $s_{\max}$.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 1 and all ablation tables report mean $\pm$ standard deviation over five seeds, where the reported standard deviation is the sample standard deviation (unbiased, $n-1$ denominator). Table captions state this convention. Welch’s t-test p -values are reported for the operating-conditions analysis (Table 3, including AVE from-scratch and CREMA-D contrasts with $p=0.0001$ and $p<0.001$). Main results (Table 1) report mean $\pm$ standard deviation over five seeds, and the cross-class MSLR comparison (Appendix B.10) follows the same five-seed protocol.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments run on a single NVIDIA RTX 4090 (24 GB VRAM). A representative CREMA-D 3-frame run (100 epochs, 6698 train and 744 test samples) takes approximately 30 minutes. Total compute for the reported results (eight datasets, ~10 methods, five seeds, plus ablations and composability sweeps) is approximately 400 GPU-hours. Preliminary hyperparameter searches that did not appear in the paper account for roughly an additional $2\times$ compute.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: This work uses only public research datasets under their standard licenses. No human subjects were involved, no data scraping or PII handling was performed, and the research does not enable surveillance, disinformation, or harmful content generation.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Positive societal impact is the primary motivator: improving balanced multimodal learning has applications in under-resourced modalities (Section 4.1 includes BraTS 2021 medical segmentation as a testbed). Negative-impact discussion: Section 5 explicitly warns that deployment should follow the operating-conditions analysis (Section 4.3.1) given the AVE from-scratch failure (-4.56 pp), the method introduces $\sim 1\%$ additional compute overhead, no new data, and no generative capability.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper introduces a training-time gradient modulation method, not a pretrained model, dataset, or generative system. No high-misuse-risk artifacts are released.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets are cited with their original papers in Section 4 and Appendix B: CREMA-D [6] (CC-BY 4.0), AVE [26], Kinetics-Sounds [2], CMU-MOSI [40], CMU-MOSEI [41], Twitter15, Sarcasm, and BraTS 2021 [3]. All datasets are used under their published research terms. ResNet-18 ImageNet-pretrained weights are obtained from torchvision (BSD-3-Clause). Baseline implementations are referenced in Section 4.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets

1050 has curated licenses for some datasets. Their licensing guide can help determine the
 1051 license of a dataset.

- 1052 • For existing datasets that are re-packaged, both the original license and the license of
 1053 the derived asset (if it has changed) should be provided.
- 1054 • If this information is not available online, the authors are encouraged to reach out to
 1055 the asset’s creators.

1056 **13. New assets**

1057 Question: Are new assets introduced in the paper well documented and is the documentation
 1058 provided alongside the assets?

1059 Answer: [N/A]

1060 Justification: No new datasets or pretrained models are released. Code will be released
 1061 upon acceptance with documentation, including a README, configuration files, and sweep
 1062 scripts.

1063 Guidelines:

- 1064 • The answer [N/A] means that the paper does not release new assets.
- 1065 • Researchers should communicate the details of the dataset/code/model as part of their
 1066 submissions via structured templates. This includes details about training, license,
 1067 limitations, etc.
- 1068 • The paper should discuss whether and how consent was obtained from people whose
 1069 asset is used.
- 1070 • At submission time, remember to anonymize your assets (if applicable). You can either
 1071 create an anonymized URL or include an anonymized zip file.

1072 **14. Crowdsourcing and research with human subjects**

1073 Question: For crowdsourcing experiments and research with human subjects, does the paper
 1074 include the full text of instructions given to participants and screenshots, if applicable, as
 1075 well as details about compensation (if any)?

1076 Answer: [N/A]

1077 Justification: The paper involves no crowdsourcing and no direct human-subjects research.
 1078 The public datasets used are independently curated.

1079 Guidelines:

- 1080 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
 1081 with human subjects.
- 1082 • Including this information in the supplemental material is fine, but if the main contribu-
 1083 tion of the paper involves human subjects, then as much detail as possible should be
 1084 included in the main paper.
- 1085 • According to the NeurIPS Code of Ethics, workers involved in data collection, curation,
 1086 or other labor should be paid at least the minimum wage in the country of the data
 1087 collector.

1088 **15. Institutional review board (IRB) approvals or equivalent for research with human
 1089 subjects**

1090 Question: Does the paper describe potential risks incurred by study participants, whether
 1091 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
 1092 approvals (or an equivalent approval/review based on the requirements of your country or
 1093 institution) were obtained?

1094 Answer: [N/A]

1095 Justification: No new human-subjects research was conducted. The datasets used are pre-
 1096 existing public benchmarks. BraTS 2021 consists of de-identified medical imaging released
 1097 under its published research license.

1098 Guidelines:

- 1099 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
 1100 with human subjects.

- 1101 • Depending on the country in which research is conducted, IRB approval (or equivalent)
1102 may be required for any human subjects research. If you obtained IRB approval, you
1103 should clearly state this in the paper.
- 1104 • We recognize that the procedures for this may vary significantly between institutions
1105 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
1106 guidelines for their institution.
- 1107 • For initial submissions, do not include any information that would break anonymity (if
1108 applicable), such as the institution conducting the review.

1109 16. Declaration of LLM usage

1110 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1111 non-standard component of the core methods in this research? Note that if the LLM is used
1112 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1113 scientific rigor, or originality of the research, declaration is not required.

1114 Answer: [N/A]

1115 Justification: LLMs are not a component of the core methodology.

1116 Guidelines:

- 1117 • The answer [N/A] means that the core method development in this research does not
1118 involve LLMs as any important, original, or non-standard components.
- 1119 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
1120 be described.